

The maximum length of a chess game under the FIDE Laws is 17,697 plies

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Abstract

Under the FIDE Laws of Chess, which end a game automatically on a fivefold repetition or when 75 moves by each player pass without a pawn move or a capture, the length of a chess game is bounded. Explicit games of 17,697 plies ($8,848\frac{1}{2}$ moves) had been published, and a construction-based count allowed 17,699 plies; to our knowledge, no published argument established that ceiling over all games or excluded the lengths 17,698 and 17,699. We prove that the maximum length of a chess game is exactly 17,697 plies. The upper bound is elementary: a pawn move or a capture resets the 75-move counter, these moves cut a game into at most 118 segments, a segment holds at most 150 plies, and one further ply is lost each time the colour making the segment-ending moves changes. The heart of the proof shows that a game attaining all 118 segments must change that colour at least three times, so that no game exceeds $150 \cdot 118 - 3 = 17,697$ plies; the equality analysis further forces every maximum-length game to promote all sixteen pawns, fill every segment, and change colour exactly three times.

1 Introduction and prior work

How long can a game of chess be? Under earlier versions of the Laws the question had no finite answer: the draw by threefold repetition and the 50-move rule must be *claimed* by a player, so two cooperating players could continue indefinitely. The unboundedness is classical: Euwe [1] exhibited endless play under the repetition rules of his day by rediscovering the cube-free Thue–Morse sequence, an analysis taken up in symbolic dynamics by Morse and Hedlund [2]. This changed on 1 July 2014, when FIDE introduced two automatic termination rules [3], retained in the current Laws [4]: the game is drawn, without any claim, when the same position occurs five times (Article 9.6.1), or when 75 consecutive moves by each player have been made without the movement of any pawn and without any capture (Article 9.6.2)¹ — with the proviso that a checkmate delivered by the final move takes precedence. The clock rules have teeth: the seven-piece endgame tablebases [6]

¹The bound of 75 has a history of its own: during the twentieth century the *claimable* 50-move limit was repeatedly extended for particular endings — at one stage to exactly 75 moves for six listed endgames — before all exceptions were abolished; see the entry “fifty-move law” of [5]. The automatic rule of 2014 is a separate, claim-independent mechanism.

contain forced checkmates more than five hundred moves long [7], wins that Article 9.6.2 erases. Under these Laws the supremum of game lengths is finite, and computing it exactly becomes a well-posed combinatorial problem — an extremal companion to the older quantitative questions about chess: Shannon’s estimates of the number of games [8], the counting and bounding of positions [9, 10, 11], the complexity of the generalised game [12].

The question is older than the automatic rules. For the claim-based 50-move setting — under the assumption that the claimable draws are in fact invoked — the answer was worked out by chess problemists over half a century: Dawson published the limit of 5,900 moves in 1911 [13], maximum-length problems by Sunyer and Høeg brought the attainable value down to 5,898, and Fabel’s finer analysis of 1947 settled the claim-based maximum at $5,898\frac{1}{2}$ moves [14]; Britten and later Roycroft treated the variant in which the two bare kings play on, as reported in [15]. Heimo’s bilingual account [16] re-examined the claim-based maximum under the Laws of 2004 — finding 5,898 whole moves, an exchange over the dead-position article adjusting his final schedule — and closed by observing that with an odd bound “such as 75” in place of 50 the answer would become White’s 8,849th move: 17,697 plies, ten years before the 75-move rule existed. The note sketches a corresponding construction, but for a hypothetical claim-based rule; we have not found in it an argument that every legal game must lose at least three plies, and the automatic termination rules under which the question becomes unconditional did not yet exist. Theorem 5.1 supplies that missing universal upper bound. We follow the account in the mathematical-chess collection of Bonsdorff, Fabel and Riihimaa [15], where the original sources are cited and the count of 118 draw-postponing opportunities — 96 pawn moves plus 22 captures not already among them — is displayed explicitly. That count is the one every later analysis, this paper’s included, rests on; what the automatic rules changed is the arithmetic built on it.

For the modern automatic rules the value itself is also not new, and its ingredients were assembled in public. The count of 118 reappeared on the Chess.com forums in 2011, together with the parity correction for switching colour — then taken to cost two full moves, via four switches [17]; Falkenbach found a three-switch plan for the 50-move setting in 2012 [17]; and within a year of the 75-move rule taking effect, Tatzelwurm and Deedlit redid the arithmetic under it, Deedlit exhibiting a three-switch schedule and putting its length at 17,697 plies [18]. Labelle [19] collected and sharpened this in 2015: counting the 118 *delay plies*, crediting the schedules, generating a complete 17,697-ply game, and stating, as a belief supported by the constructions, that the maximum is $118 \times 75.0 - 3 \times 0.5 = 8,848\frac{1}{2}$ moves; his page also records the claim-based value $5,898\frac{1}{2}$, agreeing with Fabel. Murphy [20] later published a separate, hand-designed complete game of the same length — the *reference game* verified in Appendix A — together with a construction-based count allowing 17,699 plies, but not a proof that this ceiling holds for arbitrary games, and posed the exact maximum as an open question: “We clearly need at least one switch ... but is it possible to do it with only two? If not, can this lower bound be proved?” Tromp’s reference note [21] records the value, and Bolton’s talk [22] presents the construction. The explicit games of [19] and [20] establish the lower bound 17,697; no published argument excludes 17,698 or 17,699, and to our knowledge no proof of optimality has appeared. This paper supplies one:

Main result. *Under the usual rules of chess — checkmate, stalemate, dead positions — together with the automatic fivefold-repetition and 75-move draws in force since 2014, no legal game lasts more than 17,697 plies (Theorem 5.1); the bound is attained by the published games, so the maximum length of a chess game is exactly 17,697 plies (Corollary 5.3).*

In the notation of the On-Line Encyclopedia of Integer Sequences, the result says precisely that the game-counting sequence A048987 [23] has finite support, with last nonzero term $a(17697)$.

Relative to the prior analyses the new content lies in the upper bound, and it is twofold. First, the count of 118 must be turned into a bound over *all* games: a game need not promote every pawn, so the deduction of eight overlapping moves cannot be read off the intended construction and must be derived — Proposition 3.6 couples the pawn-move and overlap counts through the number of resolved files, giving $P - O \leq 88$ for every game — and the thirtieth capture interacts with the end of the game (Lemma 3.7). Second, and this is the bulk of the paper, the three half-move losses are proved unavoidable: no ordering of the critical moves of any 118-segment game achieves fewer than three colour changes (Theorem 4.5). The equality analysis comes free and is recorded as Corollary 5.2: a maximum-length game promotes all sixteen pawns, fills every segment to its target length, and changes colour exactly three times — always in the same order, its critical-move and endpoint patterns both being BWBW.

The shape of the proof. A *critical move* — a pawn move or a capture — resets the 75-move counter and is irreversible. Critical moves cut a game into *segments*, each holding at most 149 reversible plies plus the critical move that ends it, so a game with K segments has at most $150K$ plies; a parity argument lowers this bound by one further ply each time the *colour* making the segment-ending moves changes, giving $L \leq 150K - S$ (Lemma 2.4). Counting pawn moves and captures yields $K \leq 118$ (Theorem 3.8), and $150 \cdot 118 - 1 = 17,699$ is the ceiling of [20]. The remaining two plies are removed in Section 4: a game attaining $K = 118$ is so constrained — each of the sixteen pawns must make six pawn moves, every one advancing a single rank, and promote, and at least 29 captures must occur — that its critical moves cannot be scheduled with fewer than three colour changes (Theorem 4.5). Four of the six candidate patterns are eliminated by counting alone; the two three-block patterns are eliminated by a lemma about pawns: *while the enemy’s pawns all stand on their home rank, an advancing pawn stops four moves short of promotion* (Lemma 4.3). Since $150 \cdot 117 = 17,550 < 17,697$, games with $K \leq 117$ are shorter still, and the theorem follows.

Nothing in the bound depends on how the game ends, which is why one argument covers checkmate, the 75-move rule, fivefold repetition, stalemate and *dead positions* alike — the last being the Laws’ own term for a position from which no sequence of legal moves can checkmate either side; the game is drawn the moment one arises (Article 5.2.2 [4]). That article has a literature of its own: it created the *dead reckoning* genre of retrograde-analysis problems [24, 25], has its own article in the codex of chess composition [26], and deciding it algorithmically is a published problem in its own right [27]. The proof below uses it exactly once, for king against king (Lemma 3.7).

Computer verification. The proof is a hand argument and is meant to be checked by hand; independently of it, every finite case analysis in the text was exhausted by computer, the reference game was verified ply by ply (Appendix A), and the central case analysis of Section 4 was cross-checked by two additional implementations (Appendix B). The companion repository [28] packages these checks, with a claims-to-code map (CLAIMS.md). Computation verifies the known construction and cross-checks the proof; it does not replace it.

2 Setting, critical moves, and the segment decomposition

2.1 The rules assumed

A *game* is a sequence of legal moves from the standard initial position, played until a rule of the FIDE Laws [4] ends it. We write L for its length in *plies* — half-moves, single moves by one player — and number plies from 1; White plays the odd plies. The game-ending rules relevant here are:

1. *Checkmate* (Art. 5.1.1) and *stalemate* (Art. 5.2.1) end the game on the move that produces them.
2. *Dead position* (Art. 5.2.2): the game is drawn the moment a position arises from which no sequence of legal moves can lead to a checkmate by either side.
3. *Fifefold repetition* (Art. 9.6.1): the game is drawn, with no claim required, when the same position (same pieces on the same squares, same side to move, same castling and en passant possibilities, as specified in Art. 9.2) has occurred five times.²
4. *Seventy-five-move rule* (Art. 9.6.2): the game is drawn, with no claim required, when at least 75 consecutive moves by each player — that is, 150 consecutive plies — have been played without any pawn move and without any capture. If the ply completing the 150 delivers checkmate, the checkmate stands.

Draws by agreement, resignations, and the *claimable* draws (threefold repetition, Art. 9.2; the 50-move rule, Art. 9.3) are voluntary: players maximising the length of a game simply never invoke them, so they play no role in the maximisation. This is the setting of [20].

It is convenient to track the *halfmove clock* $c(t)$: the number of plies played since the most recent pawn move or capture (or since the start of the game, if there has been none). Article 9.6.2 says precisely that the game ends when c reaches 150, the ply that brings it there being the last of the game, and being scored as mate if it mates.

2.2 Critical moves and segments

Definition 2.1 (Critical move). A *critical move* is a pawn move or a capture. (En passant is both; a promotion is a pawn move; castling is neither.) A *quiet* move is a move that is not critical.

²The 2014 formulation required the five occurrences to be consecutive; the requirement was removed in the 2017 revision [29]. The difference is immaterial here: the upper bound follows from the automatic 75-move rule alone, and the verified reference game never lets any position occur even three times (Appendix A).

Critical moves are exactly the moves that reset the halfmove clock, and they are irreversible: a pawn cannot retreat, and a captured piece never returns. We keep the term of [20], where the problem was posed: “Let’s call an irreversible move that resets the 75-move counter a *critical move*.” Labelle [19] calls the same plies *delay plies*; the usage is unrelated to the colloquial sense of a tactically critical move.

Definition 2.2 (Segments, endpoints, K , T). Let $t_1 < t_2 < \dots < t_m$ be the plies of a game’s critical moves. Set $T = 1$ if quiet plies follow the last critical move (or if the game has moves but no critical move at all), and $T = 0$ otherwise. The game’s *endpoints* are

$$e_i = t_i \ (1 \leq i \leq m), \quad e_{m+1} = L \text{ if } T = 1,$$

together with the virtual endpoint $e_0 = 0$. The number of *segments* is $K = m + T$; segment i consists of plies $e_{i-1} + 1, \dots, e_i$. Every ply lies in exactly one segment, and $e_K = L$.

Definition 2.3 (Switches S). The *colour* of an endpoint e_i ($i \geq 1$) is the colour of the player who played ply e_i — White if e_i is odd, Black if even. By convention the virtual endpoint $e_0 = 0$ is *Black* (it has the parity of a Black ply, and the game opens as if Black had just made a critical move). The switch count is

$$S = \#\{i \in \{1, \dots, K\} : \text{colour}(e_i) \neq \text{colour}(e_{i-1})\}.$$

Lemma 2.4 (Segment decomposition). *Every game satisfies*

$$L = 150K - S - \sum_{i=1}^K \delta_i \quad \text{with every } \delta_i \geq 0,$$

where δ_i is the shortfall of segment i from its target length (150 if endpoints e_{i-1}, e_i have the same colour, 149 otherwise). In particular $L \leq 150K - S$.

Proof. Fix i and consider the segment plies $e_{i-1} + 1, \dots, e_i$. All of them except possibly the last are quiet (t_i is the next critical ply after t_{i-1} ; the terminal segment is entirely quiet). Suppose $e_i - e_{i-1} \geq 151$. Then the 150 plies $e_{i-1} + 1, \dots, e_{i-1} + 150$ are all quiet, so after ply $e_{i-1} + 150$ the halfmove clock reads 150 and the game has ended there by Article 9.6.2 (as a draw, or as the mate that takes precedence — either way it has ended), contradicting the existence of ply $e_{i-1} + 151 \leq e_i \leq L$. Hence $e_i - e_{i-1} \leq 150$.

Plies alternate colours, so $e_i - e_{i-1}$ is even exactly when e_i and e_{i-1} have the same colour; this is where the convention $\text{colour}(e_0) = \text{Black}$ is used, 0 having the parity of a Black ply. A segment whose endpoints differ in colour therefore has odd length, and being ≤ 150 , has length ≤ 149 . Summing the segment lengths telescopes to $e_K - e_0 = L$ and yields the identity. $\square$

Remark 2.5. The terminal quiet segment genuinely can reach 150 plies: if the game ends by the 75-move rule, the drawing ply is the 150th quiet ply after the last critical move and it *is played*. Its parity then forces the terminal endpoint to have the same colour as the previous one, consistent with the accounting above. A terminal segment ending earlier — by a quiet checkmate, a stalemate, a fivefold repetition or a dead position — is only shorter.

Counting critical moves by inclusion–exclusion, with P the number of pawn moves, C the number of captures and O the number of *overlaps* (moves that are both, i.e. diagonal pawn captures, en passant included),

$$K = (P - O) + (C + T). \quad (1)$$

The next two sections bound the two halves of (1) separately. Neither half refers to how the game ends.

3 At most 118 critical segments

Throughout, we count over the 32 *pieces* that begin the game — each one followed through the whole game, keeping its identity under promotion — rather than over piece types; “piece” is used inclusively here, so pawns are pieces too. A piece that begins as a pawn is still that same piece after promoting.

Lemma 3.1 (Pawn basics). *Each pawn makes at most 6 pawn moves, so $P \leq 96$. A pawn that makes exactly 6 pawn moves advances exactly one rank with each of them and promotes.*

Proof. Every pawn move advances the pawn at least one rank toward promotion, and there are six ranks between its home rank and the promotion rank. A pawn making six moves therefore advances exactly one rank per move and finishes on the promotion rank, where promotion is mandatory. $\square$

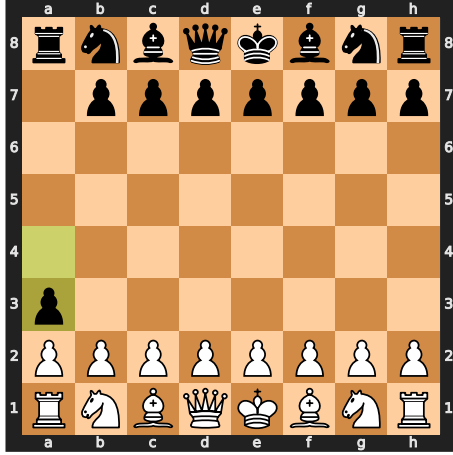
3.1 Origin pairs and the bound $P - O \leq 88$

Definition 3.2 (Origin pair, resolved file). The two pawns that begin on file i (one White, one Black) are that file’s *origin pair*. File i is *resolved* if at least one of those two pawns makes a diagonal pawn move at some point in the game; otherwise it is *unresolved*. Write f for the number of resolved files.

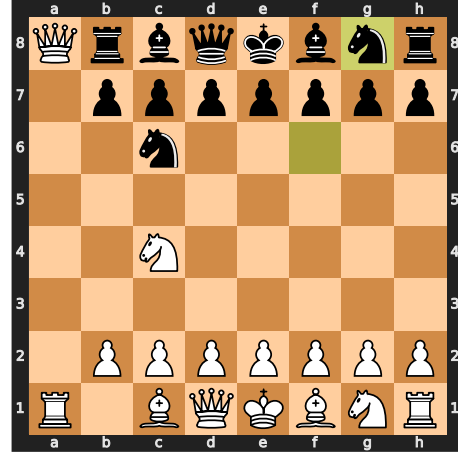
A diagonal pawn move is the only way a pawn changes file, and it is always a capture; so both pawns of an unresolved pair spend their entire lives on file i .

Lemma 3.3 (Unresolved pair bound). *The two pawns of an unresolved origin pair make at most 10 pawn moves combined, and 10 is attained in a legal game.*

Proof. Both pawns stay on file i . While both are on the board, the White pawn is strictly below the Black one: neither can capture the other (a capture would be diagonal), and neither can pass the other, since a single push onto an occupied square is illegal and a double push may not jump over an occupied square. So if White’s pawn has advanced a ranks and Black’s b ranks, they stand on ranks $2 + a < 7 - b$, giving $a + b \leq 4$; and since each pawn move advances at least one rank, the two pawns have made at most 4 combined moves while both are on the board. In particular neither pawn can promote while the other is on the board, so at most one of the pair ever promotes, and only after the other has been captured (necessarily by some other piece).



(a)



(b)

Figure 1: The unresolved origin pair attaining ten combined pawn moves (Lemma 3.3). (a) After 4...a3: Black's a-pawn has made four moves and is captured on the next move by the b1-knight. (b) The final position: White's a-pawn has walked the file and promoted, six further moves. Neither pawn ever left the a-file.

When one pawn of the pair leaves the board its total contribution is frozen at some $d \leq 4$, and the survivor's lifetime total is at most 6 by Lemma 3.1, for a combined total of at most $d + 6 \leq 10$.

Ten is attained: in

1. Nf3 a6 2. Ng1 a5 3. Nf3 a4 4. Ng1 a3 5. Nxa3 Nc6 6. Nc4 Rb8 7. a3 Nf6 8. a4 Ng8
9. a5 Nf6 10. a6 Ng8 11. a7 Nf6 12. a8=Q Ng8

Black's a-pawn walks $a7$ – $a3$ in four moves and is captured there by the knight; White's a-pawn then walks the whole file and promotes, six more. Neither pawn ever moved diagonally, so the a-file is unresolved throughout. The sequence is a legal prefix rather than a completed game, but in any legal completion neither count changes — the survivor has promoted and the other has been captured, so the pair makes no further pawn move and the file stays unresolved — so a completed legal game attains ten as well. Figure 1 shows the position after Black's fourth pawn move and the final position of the sequence. $\square$

Remark 3.4. We record a caution for anyone reproducing this bound. The first observation of the proof — “two facing pawns cannot pass, so the pair makes at most 4 combined pawn moves” — holds only *while both pawns are on the board*; taken as an upper bound for the whole game it is false, since capturing one pawn frees the survivor to advance.

Lemma 3.5 (Overlap floor). $O \geq f$.

Proof. For each resolved file i choose one diagonal pawn move made by a pawn whose *origin* file is i ; one exists by definition, and it is a capture. A pawn has one origin file, so the chosen moves of distinct resolved files have distinct movers and are distinct moves, each of them an overlap. $\square$

Proposition 3.6. $P - O \leq 88$, and equality forces $f = 8$, $P = 96$ and $O = 8$.

Proof. The eight origin pairs partition the sixteen pawns, so summing Lemma 3.3 over unresolved pairs and the trivial bound 12 over resolved pairs, and combining with Lemma 3.1,

$$P \leq \min(96, 10(8 - f) + 12f) = \min(96, 80 + 2f).$$

With Lemma 3.5, $P - O \leq \min(96, 80 + 2f) - f$. For $f \leq 7$ this is at most $80 + 2f - f = 80 + f \leq 87$; for $f = 8$ it is $96 - 8 = 88$. The maximum over f is therefore 88, attained only at $f = 8$ with $P = 96$ and $O = f = 8$. $\square$

Maximising over f is what makes Proposition 3.6 a statement about all games rather than about games with eight overlaps; asserting $O \geq 8$ outright would be circular, being true only *because* $K = 118$ forces every file to be resolved.

3.2 The bound $C + T \leq 30$

Lemma 3.7. $C + T \leq 30$.

Proof. Only the 30 non-king pieces can ever be captured, so $C \leq 30$. If $C \leq 29$ then, since $T \leq 1$, we get $C + T \leq 30$. If $C = 30$, the 30th capture leaves the two kings alone on the board. King versus king is a dead position — no sequence of legal moves mates — so by Article 5.2.2 the game ends on that very ply; no quiet ply follows the last critical move and $T = 0$, giving $C + T = 30$. $\square$

This is the only place any dead-position reasoning enters the upper bound, and king versus king is the one case where deadness is beyond dispute. The two ways of reaching 30 are worth displaying, because they explain why draws do not outlast checkmates:

$$\underbrace{96 + 30 - 8 + 0}_{\text{all 30 captured, no closing segment}} = \underbrace{96 + 29 - 8 + 1}_{\text{29 captured, one closing segment}} = 118.$$

3.3 The bound $K \leq 118$ and its equality case

Theorem 3.8. Every game has $K \leq 118$. Moreover a game with $K = 118$ satisfies

$$f = 8, \quad P = 96, \quad O = 8, \quad C + T = 30,$$

so all sixteen pawns make six pawn moves each — every one advancing the pawn by exactly one rank — and promote, and $C \geq 29$.

Proof. By (1), Proposition 3.6 and Lemma 3.7,

$$K = (P - O) + (C + T) \leq 88 + 30 = 118.$$

Equality forces $P - O = 88$ and $C + T = 30$; the former pins $(f, P, O) = (8, 96, 8)$ by the equality case of Proposition 3.6, and $P = 96$ makes every pawn's count exactly 6, whence Lemma 3.1 applies to each. Finally $T \leq 1$ gives $C \geq 29$. $\square$

Remark 3.9 (The ceiling of 17,699). Theorem 3.8 already yields the ceiling stated in [20], now as a bound over all games. A game with $K \leq 117$ has $L \leq 150 \cdot 117 = 17,550$ by Lemma 2.4. A game with $K = 118$ has $P = 96$, so both sides move pawns; a pawn move is a critical move made by the pawn's owner, so endpoints of both colours occur. Since the virtual endpoint e_0 is Black by convention and some endpoint is White, the endpoint colour sequence changes at least once: $S \geq 1$, and $L \leq 150 \cdot 118 - 1 = 17,699$. The next section shows that at $K = 118$ the colour must in fact change three times, which removes the last two plies.

4 Games with 118 segments have at least four critical-move blocks

Throughout this section, fix a game with $K = 118$; by Theorem 3.8 it has $P = 96$ — *every pawn makes exactly six pawn moves* — and $C \geq 29$. These two facts do all the work. We emphasise at the outset that $S \geq 3$ is claimed *only* under $K = 118$; it is false in general (Remark 4.6).

4.1 Blocks and patterns

The critical moves of a game, in order, form a sequence of coloured events (the colour of a pawn move is the pawn's owner; the colour of a capture is the capturer, i.e. the side that does *not* own the piece captured). The *pattern* of any finite sequence of coloured events is obtained by grouping maximal runs of equal colour into *blocks* and listing the block colours; the pattern of the critical-move sequence is the game's *critical-move pattern*, and the pattern of the endpoint sequence is its *endpoint pattern*. Here the endpoint sequence means $(e_1, \dots, e_K)$ of Definition 2.2: the virtual endpoint e_0 is *not* part of the endpoint pattern; it is prepended only when computing the switch count S . Counting that virtual Black endpoint, a game whose endpoint pattern consists of n blocks has $S = n - 1$ if the pattern opens with Black and $S = n$ if it opens with White. And since the endpoint sequence is the critical-move sequence with, when $T = 1$, one element appended — which either extends the last block or starts a new one — the endpoint pattern has at least as many blocks as the critical-move pattern.

A game with $K = 118$ has at least 117 critical moves, so its critical-move pattern is nonempty. We refute every nonempty alternating pattern of at most three blocks — there are exactly six:

$$B, \quad W, \quad BW, \quad WB, \quad BWB, \quad WBW.$$

Each side has 15 capturable pieces, of which exactly 7 (queen, two rooks, two bishops, two knights) did not begin as pawns; the count below is over pieces in the sense of Section 3, so a piece that began as a pawn and is captured after promoting still counts against the pawn origins.

4.2 Four patterns are eliminated by counting

Proposition 4.1 (Single block). *Patterns B and W are impossible at $K = 118$.*

Proof. A pawn move is a critical move of the pawn's own colour. In a single-block game only one colour makes critical moves, so only its eight pawns ever move: $P \leq 48 < 96$. $\square$

Proposition 4.2 (Two blocks). *Patterns BW and WB are impossible at $K = 118$.*

Proof. Write the pattern (X, Y) : every critical move of X precedes every critical move of Y . Write C_X and C_Y for the numbers of captures made by X and by Y , so that $C_X + C_Y = C$. Since Y captures only X 's pieces, $C_Y \leq 15$, so $C_X \geq C - 15 \geq 29 - 15 = 14$: the colour moving first makes at least 14 captures, all of them before Y has made a single critical move. Each piece so captured is a Y piece that, being off the board before every Y critical move, never made a pawn move. At most 7 of the captured pieces did not begin as pawns, so at least 7 of Y 's pieces that began as pawns make no pawn move at all, and

$$P \leq 96 - 7 \cdot 6 = 54 < 96. \quad \square$$

Note that Y may well have made *quiet* moves in the meantime; the argument only needs its critical moves to come later.

4.3 The home-rank lemma

The three-block patterns are not eliminated by counting alone; the following lemma closes both.

Lemma 4.3 (Home-rank lemma). *Fix colours A (attacker) and D (defender), and consider any position reachable from the initial position by a sequence of moves satisfying*

(R1) *every move by D is quiet, and*

(R2) *no move by A captures a D pawn.*

Then every square of D 's pawn home rank still holds the D pawn that started there; no A pawn ever stands on that rank; and no A pawn has made more than four pawn moves.

Proof. We use only the following elementary facts about legal chess moves, labelled for reference: [H0] the initial position has eight pawns on each home rank; [H1] a legal move starts on a square holding a piece of the side to move; [H2] a move onto an occupied square is a capture of an enemy piece standing there; [H3] castling changes only back-rank squares; [H4] an en passant capture removes a pawn from rank 4 or rank 5; [H5] a move changes the occupancy of no squares other than its origin and destination, the castling rook's two squares, and the square of the pawn captured en passant; [H6] a pawn move advances the pawn exactly one rank, or exactly two from its own home rank only.

Part 1: D 's home rank never changes. Induction on moves, the base case being [H0]. Let the invariant hold and let m be a move permitted by (R1), (R2). By [H5] it suffices to rule out each way m could touch a home-rank square. m cannot *start* there: the square holds a D pawn, only D may move it [H1], and that would be a pawn move, excluded by (R1). m cannot *end* there: the square is occupied, so the move would capture an enemy piece standing there [H2]; D capturing is excluded by (R1), and the occupant is a D pawn, so A capturing it is excluded by (R2). Castling touches only back ranks [H3]. En passant

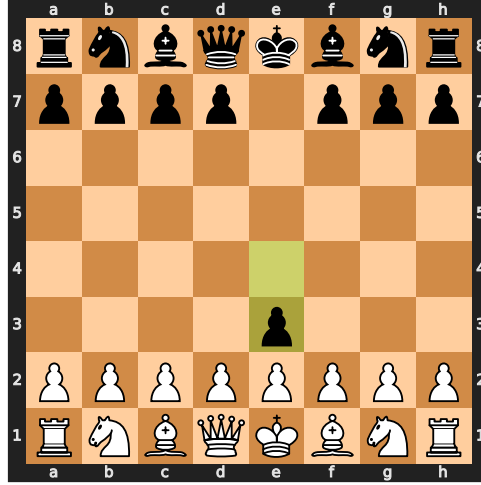


Figure 2: The home-rank lemma (Lemma 4.3), minimal instance. After 1. Nf3 e6 2. Ng1 e5 3. Nf3 e4 4. Ng1 e3, Black’s pawn has made four moves and is confined: the square e2 holds a pawn that only White may move, which would violate (R1), and the captures on d2 and f2 are exactly what (R2) forbids.

removes a pawn from rank 4 or 5 [H4], not from a home rank (and the pawn captured would in any case have just made a double push — for a D pawn a pawn move, excluded by (R1)).

Part 2: no A pawn reaches D ’s home rank. Standing there requires a move ending there, ruled out above.

Part 3: at most four moves. By [H6] a pawn never moves backwards and double-pushes only from its own home rank. By Part 2 an A pawn never occupies D ’s home rank; nor the rank beyond it, which could only be reached by a single step from D ’s home rank or a double push from a rank that is not the pawn’s home rank. Its range is therefore the four ranks between its own home rank and D ’s, and each pawn move spends at least one of them: at most four moves. (Four is attained, by four single pushes.) $\square$

Figure 2 shows a minimal instance of the confinement.

Proposition 4.4 (Three blocks). *Neither BWB nor WBW is possible as the critical-move pattern of a game with $K = 118$.*

Proof. Write either pattern as (X, Y, X) , where X and Y are the two colours, and write C_X, C_Y for the numbers of captures made by X and by Y ; thus every critical move of Y lies in the middle block. Let τ be the ply of Y ’s first critical move, and call the plies before τ the *first phase*.

Since X can capture only the 15 capturable pieces of Y , we have $C_X \leq 15$, so

$$C_Y = C - C_X \geq 29 - 15 = 14.$$

All these captures are critical moves of Y and therefore occur in the middle block. At most seven of the pieces captured by Y did not begin as pawns, so at least seven of them are X -pieces that began as pawns.

Fix one such piece p . By Theorem 3.8 it makes exactly six pawn moves, all before its capture. Every pawn move of p is a critical move of X , and a critical move of X at a ply $> \tau$ belongs to the final X -block, hence comes after *every* critical move of Y — in particular after the move on which p is captured, which is impossible for a piece no longer on the board. So all six of its pawn moves occur in the first phase.

Now apply Lemma 4.3 to the first phase with $A = X$ and $D = Y$. (R1) holds by the definition of τ : every move by Y before τ is quiet. For (R2), suppose a move by X in the first phase captured a Y pawn. That pawn would be off the board before Y 's first critical move; since each of its pawn moves would itself be a critical move of Y , it would make no pawn move at all, contradicting $P = 96$. So (R2) holds as well. The lemma bounds every X pawn by four pawn moves in the first phase — but each of the seven pieces fixed above needs six there. Counting only the shortfall of two for each,

$$P \leq 96 - 7 \cdot (6 - 4) = 82 < 96. \quad \square$$

The only escape the home-rank lemma allows — capturing a way through the home rank — costs the captured pawn all six of its moves, which Theorem 3.8 cannot spare. That is the entire content of clause (R2) above.

Theorem 4.5 (Four blocks). *Every game with $K = 118$ has a critical-move pattern of at least four blocks; hence its endpoint pattern also has at least four blocks, and $S \geq 3$.*

Proof. The critical-move pattern is a nonempty alternating pattern, and the six such patterns of at most three blocks are refuted by Propositions 4.1, 4.2 and 4.4; so it has at least four blocks, and by the appending observation of Section 4 so does the endpoint pattern. A game whose endpoint pattern has $n \geq 4$ blocks has $S \geq n - 1 \geq 3$. $\square$

Remark 4.6. $S \geq 3$ is conditional on $K = 118$, and only on it: every step above runs on $P = 96$ and $C \geq 29$. It is plainly false in general. The 16-ply game 1.Nf3 Nf6 2.Ng1 Ng8 repeated four times is legal and ends in fivefold repetition of the initial position; it has no pawn move and no capture, so $K = 1$ (the single closing segment), and its one endpoint is a Black ply, matching the virtual Black opening: $S = 0$. This costs nothing in Section 5: a game with $K \leq 117$ forfeits at least 150 plies, more than any switch count could recover.

5 The maximum length

Theorem 5.1 (Upper bound). *No legal chess game lasts more than 17,697 plies.*

Proof. Let a game have K segments. If $K \leq 117$ then by Lemma 2.4 with the trivial $S \geq 0$,

$$L \leq 150 \cdot 117 = 17,550 < 17,697.$$

If $K = 118$ — the largest value Theorem 3.8 permits — then Theorem 4.5 gives $S \geq 3$, so

$$L \leq 150 \cdot 118 - 3 = 17,697. \quad \square$$

The two cases must be kept separate, because $S \geq 3$ is not available at $K \leq 117$ (Remark 4.6); it is also not needed there, since one whole segment outweighs any possible switch count.

The equality analysis costs nothing further, and it determines a maximum-length game rather precisely:

Corollary 5.2 (Structure of maximum-length games). *A game of length 17,697 satisfies*

$$K = 118, \quad S = 3, \quad \delta_i = 0 \text{ for every } i,$$

and therefore, by the equality case of Theorem 3.8,

$$f = 8, \quad P = 96, \quad O = 8, \quad C + T = 30.$$

In words: all sixteen pawns make six pawn moves each — every one advancing the pawn by exactly one rank — and promote, at least 29 captures occur, every one of the 118 segments attains its target length (150 plies, or 149 across a colour change), and both the critical-move pattern and the endpoint pattern are BWBW: Black makes the first critical move, and the final endpoint is White's.

Proof. By Theorem 3.8, $K \leq 118$; if $K \leq 117$ the game has at most 17,550 plies, so $K = 118$, and Theorem 4.5 gives $S \geq 3$. The decomposition of Lemma 2.4 then reads

$$17,697 = 150 \cdot 118 - S - \sum_{i=1}^{118} \delta_i \leq 150 \cdot 118 - 3 = 17,697,$$

with equality throughout: $S = 3$ and every $\delta_i = 0$. The equality case of Theorem 3.8 supplies f , P , O and $C + T$, and Lemma 3.1 the behaviour of each pawn.

For the patterns, Theorem 4.5 gives at least four blocks in the critical-move pattern. Since $S = 3$, there can be no more than four: each change between consecutive critical-move blocks is an adjacent unequal pair of the endpoint sequence, already counted by S . Thus the critical-move pattern has exactly four blocks. It cannot open with White: the initial comparison with the virtual Black endpoint e_0 would then add one switch to the three changes between the four blocks, giving $S \geq 4$. Hence the critical-move pattern is BWBW. Finally, a terminal endpoint, if present, cannot start a new block, since that would add a fourth switch; therefore the endpoint pattern is also BWBW. $\square$

Corollary 5.3 (Exact maximum). *The maximum length of a chess game under the FIDE Laws is exactly 17,697 plies — 8,848 full moves and one final White ply.*

Proof. The bound of Theorem 5.1 is attained: Murphy's explicit game [20] is 17,697 plies long and ends in checkmate on White's 8,849th move, as independently re-verified, ply by ply, in Appendix A, where the game is also matched against Corollary 5.2. $\square$

Remark 5.4. The reference game realises Corollary 5.2 with $C = 29$ and $T = 1$: its final ply is a quiet checkmate delivered with the halfmove clock at exactly 150, the checkmate taking precedence under Article 9.6.2. Improving on such a game is impossible rather than merely hard: it fills every segment, and its three lost plies are exactly the three switches Theorem 4.5 proves unavoidable.

Acknowledgements

This work exists because Tom Murphy VII built a 17,697-ply game and asked, in the title of [20], whether it is the longest; the answer is yes. The numerical structure behind the answer — 118 clock-resetting opportunities, and constructions incurring three half-move losses — was identified years earlier by François Labelle [19] and by the contributors of the forum threads his page credits — Deedlit, Tatzelwurm and Falkenbach among them [17, 18].

We thank Alexis Langlois-Rémillard for helpful comments on an earlier draft, in particular on its chess and mathematical vocabulary.

This work was carried out independently, outside any research group and without funding.

The board diagrams were rendered with `python-chess` [30]; the piece images are by Colin M. L. Burnett [31], triple-licensed under the GFDL, BSD and GPL, and are used here under the BSD option with this attribution.

A The reference game

Among the published maximum-length games we verify the one constructed by hand in [20] — the *reference game*; Labelle’s page [19] supplies another, machine-generated. (The repository’s documentation and `make` targets call the reference game the *witness*.) The companion repository [28] contains an independent verifier for it, built on the move generation of the `python-chess` library [30] and this project’s own implementation of the termination rules of Section 2.1: it replays the game from the initial position, rejects any move that is illegal or played after the game has ended, classifies the position after every ply, and emits a per-ply trace (clock value, position multiplicity, termination state); a sequence passes only if the game ends exactly on its last move. The verified facts:

- The game is legal, 17,697 plies long, and ends in checkmate, in the final position `3k4/3Q4/4K3/8/8/8/8/8` — king and queen against king, Figure 3.
- No automatic termination fires early. The halfmove clock stays below 150 after every ply but the last; no position ever occurs even three times (the maximum multiplicity of any position over the whole game, under the identity of Article 9.2, is *two*), so fivefold repetition is never approached; no stalemate occurs.
- No position of the game is a dead position, by an argument that needs no material test: from every position, the remaining moves of the game are themselves a legal series ending in checkmate, and a position from which checkmate is reachable is precisely what Article 5.2.2 does not end. The verifier checks the legality of those suffixes.
- The final ply is a quiet checkmate played with the halfmove clock at exactly 150: the game ends on the precedence clause of Article 9.6.2. Its length is insensitive to that clause — under either reading the 17,697th ply is played and the game ends on it; the clause decides only whether the result is a win or a draw.

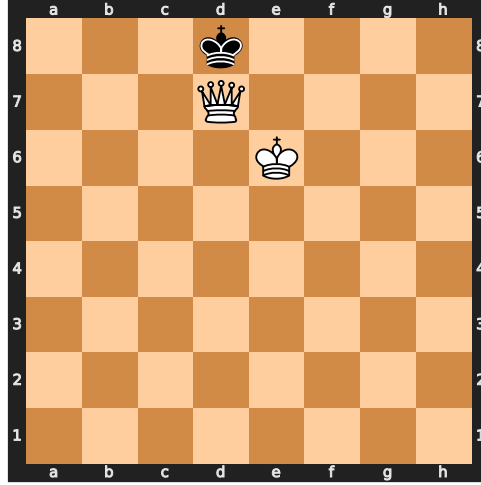


Figure 3: The final position of the reference game, after ply 17,697: checkmate, delivered with the halfmove clock at exactly 150.

- The profile is the one Corollary 5.2 forces: 117 critical moves plus a closing segment ($K = 118$), with $P = 96$ (each of the sixteen pawns makes exactly six pawn moves), $C = 29$, $O = 8$ (every file resolved), $T = 1$. The endpoint pattern is BWBW, so $S = 3$, and the decomposition of Lemma 2.4 closes exactly: 115 segments of 150 plies, 3 segments of 149 (one per switch), and $\sum_i \delta_i = 0$:

$$17,697 = 150 \cdot 118 - 3 - 0.$$

Two points of this verifier are rules of chess rather than engineering choices, and the reference game exercises both. First, the order of the termination tests: checkmate is tested before the 150th-quiet-ply test, because the final ply above is simultaneously both, and the precedence clause of Article 9.6.2 says the mate stands — a verifier that consults the clock first accepts the same 17,697 plies but scores the game a draw. (The verifier’s material-draw screen is not a decision procedure for Article 5.2.2 — for such a procedure see [27]; it is never load-bearing here, by the suffix argument above.) Second, position identity for repetition: following Article 9.2, two positions count as the same exactly when their piece placement, the side to move, the *retained* castling rights — whether each side still has the right to castle with each original rook, regardless of whether castling is legal in the current position — and the availability of a *legal* en passant capture all agree. Naive position keys fail in both directions. Comparing whole FEN strings separates positions the Laws identify: the halfmove clock and the move number play no part in identity, and the en passant target is recorded after every double push whether or not a capture is actually legal — so the fivefold rule would never fire. Comparing only piece placement and the side to move merges positions the Laws distinguish, since retained castling rights and a legal en passant capture also matter. The verifier keys its repetition table on exactly the four data above; under that identity, no position of the reference game ever occurs three times. The verifier is deliberately independent: it imports nothing else from the repository, so the judge shares no code with any producer of games.

Nor does the verification rest on a single implementation of the rules. A second verifier — one file of C99 in the companion repository, using a 0x88 mailbox board where `python-chess` uses bitboards, parsing the PGN and resolving its SAN for itself, written against the Laws without reading `python-chess` — replays the same game to the same verdict: 17,697 plies, checkmate, 118 critical segments, a per-ply trace byte-identical to the Python verifier’s across all 17,698 rows, and the identical set of legal moves at every position of the game (730,845 moves in all). Its own move generation is checked against perft node counts published outside this project and against hand-written rule cases pinned to the articles of the Laws.

B Computational cross-checks

Every finite case analysis in the text, each also proved by hand, was exhausted by computer: the unresolved-pair bound of Lemma 3.3, over a state space strictly more permissive than chess (turn order ignored, no third piece blocking the file, either pawn removable at any moment without cost); the impossibility of two facing pawns passing each other on a file; the uniqueness of the equality assignment of Theorem 3.8 (the same enumeration at $K = 117$ admits many assignments, which is what makes the pinning at 118 informative); the terminal-endpoint step of Theorem 4.5 and Corollary 5.2 — appending the terminal endpoint cannot lower the block count, and starting a new block adds a switch — over every alternating pattern of up to twelve blocks; and, for Lemma 4.3, the finite obligations [H0], [H3] and [H4]. The universally quantified obligations [H1], [H2], [H5], [H6] admit no enumeration: they are read off the Laws [4], and the repository checks that both move generators respect them, each over a large randomised corpus of positions — a check on the tooling, not on the mathematics.

To reduce the risk of an undetected error in the hand argument of Section 4, the question “does any endpoint pattern of at most three blocks admit $K = 118$?” is cross-checked by two additional computational methods: a CP-SAT constraint model [32], and a solver-free arithmetic checker that declares its own constants, imposes aggregated forms of the same conditions — each likewise a necessary condition of legality — and decides the same question by direct enumeration.

Fix an alternating endpoint pattern σ , an ending $e \in \{\text{checkmate}, \text{draw}\}$, and a target k . The model $M(\sigma, e, k)$ is a finite system of constraints over bounded integer variables: per-pawn pawn-move counts in each block of the pawn’s colour, with a per-pawn total of at most six; per-piece capture indicators, each captured piece assigned the enemy-colour block in which it is captured, a captured piece that began as a pawn making no pawn move in any later block; capture, overlap and closing-segment totals with the coupling of Proposition 3.6 imposed through a free resolved-file count; the bound of Lemma 4.3 on opening-block pawn moves, imposed only while no enemy pawn has been captured in that block; for $e = \text{checkmate}$, the requirement that the side owning the final block retain at least one non-king piece; and the requirement that the totals compose to k as in (1).

Soundness property. Let G be a legal game with $K(G) = k$ whose endpoint pattern is σ , and set e to checkmate if G ends in mate and to draw otherwise. Then the observed values of G — its per-pawn, per-block move counts, its capture blocks, and its totals —

satisfy every constraint of $M(\sigma, e, k)$. This holds because each constraint is individually a necessary condition of legality; the constraint-by-constraint argument is recorded in the repository’s model audit, and the attribution of the mating side to the final block is sound precisely because σ is a pattern of *segment endpoints* (Definition 2.2), the mate being the last endpoint. Consequently, if $M(\sigma, e, 118)$ is infeasible for both values of e , then no legal game with endpoint pattern σ attains $K = 118$. No claim is made in the converse direction: a satisfying assignment need not correspond to any legal game, and no conclusion of this paper rests on a feasibility verdict.

Coverage is finite. An alternating pattern of $n \geq 5$ blocks has $S \geq 4$ by the block count of Section 4, so the sixteen decided instances — the eight endpoint patterns of at most four blocks, under both endings — include every pattern that the refutations of Section 4 concern. CP-SAT decides each finite instance exactly; a solver timeout is surfaced as an error and never reported as infeasibility. The results, identical under both endings: all six endpoint patterns of at most three blocks are infeasible, and the model is feasible exactly for BWBW and WBWB — in agreement with Theorem 4.5, whose endpoint half asserts precisely that no endpoint pattern of at most three blocks occurs at $K = 118$. (The propositions of Section 4 refute critical-move patterns; the theorem carries the refutation over to endpoint patterns, which are what the model decides.) Removing the home-rank constraint changes exactly BWB and WBW from infeasible to feasible and leaves every other verdict unchanged, confirming that the lemma carries precisely the two cases Proposition 4.4 assigns to it and that no other refutation of Section 4 depends on it.

The soundness property cannot be established by testing, but violations of it can be exhibited: the constraint model accepts the exact assignment induced by the reference game of Appendix A, and both deciders return feasible for its endpoint pattern BWBW. Three constraints that were, at stages of this work, stronger than legality — the first of them the misreading against which Remark 3.4 warns — were each found by the constraint-by-constraint audit, repaired, and pinned by regression tests; none survives in the verdicts above, and the full record, with explicit refuting games, is in the repository.

What a reader of these checks must trust: first, the summary of the Laws in Section 2.1 — in particular that Articles 9.6.1–9.6.2 end games automatically and that mate takes precedence on the final ply; both implementations below follow this project’s reading of the Laws, so their agreement cannot vouch for the reading itself. Second, move generation. Every check above runs through `python-chess` [30] (version pinned in the repository), and the obligations [H1], [H2], [H5], [H6] are facts about chess that we verify the generators respect, not facts we derive from them; but the reference game is also re-verified by the C implementation of Appendix A, which shares no code, no board representation and no input parsing with `python-chess`, and whose own generation is checked against externally published perft counts. What remains to be trusted here is therefore not one library but the proposition that two independently implemented move generators do not fail identically at the same positions. The reference game is an artefact produced by others [20] and accepted by both verifiers, separating producer from judge. The claims-to-code map (`CLAIMS.md`), invocation commands, corpus and search statistics and the full audit record live in the companion repository [28]; the repository version corresponding to this paper is tag `v1.1.0`, archived at the DOI cited there.

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